Leveraged ETF Research Report

SOX Semiconductor Index — 1x / 2x / 3x compared over 32 years (1994-2026)

Item	Detail
Version	v1.0 (2026-10-27)
Scope	Daily-reset leveraged exposure, 1x / 2x / 3x
Data	Yahoo Finance daily, 1994-05-04 to 2026-10-23 (32.5 years)
Costs	Expense ratio 0.35% / 0.95% / 0.75%; financing at 13-week T-bill (^IRX)
Method	Synthetic daily-reset, independent recomputation

This document is an analytical report, not investment advice.

1. Executive summary

Over 32 years, the underlying index compounded at +15.87% a year. The 2x series compounded at +13.21%, and the 3x series at -3.63%. Maximum drawdowns were 86%, 100% and 100%.

Metric	1x	2x	3x
CAGR	+15.87%	+13.21%	-3.63%
Volatility	37.9%	75.9%	113.8%
Max drawdown	86.2%	99.8%	100.0%
Sharpe	0.47	0.49	0.50
Calmar	0.18	0.13	-0.04
Terminal multiple	119x	56x	0x

Three findings stand out. First, leverage raises the compounded return but not the risk-adjusted return: Sharpe ratios are essentially unchanged. Second, 3x does not beat 2x over long horizons. Third, the outcome is decided by the ratio of drift to variance, not by volatility alone.

Breakeven volatility for 3x is sqrt(2*mu/3), where mu = ln(1+CAGR). With CAGR = 15.9%, mu = 14.7% and sigma* = 31.3%. Realised volatility was 37.9%, so 3x has little theoretical room.

2. Data and method

The index history is converted into synthetic leveraged series using the daily-reset identity below. Financing is charged at the realised 13-week T-bill rate, and the expense ratio of a representative ETF is deducted daily.

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1 + R_L = 1 + L*R - (L-1)*c - f
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Two assumptions matter. The index price series excludes dividends, so a constant dividend yield is assumed; a sensitivity without dividends is shown in section 5. The series is a single historical path, not a forecast.

Table 1. Inputs

Input	Value
Price index	SOX.csv daily
Dividend assumption	1.2% per year, constant
Financing	13-week T-bill (^IRX), daily
Expense ratio	0.35% / 0.95% / 0.75%
Rebalance	daily

3. Results

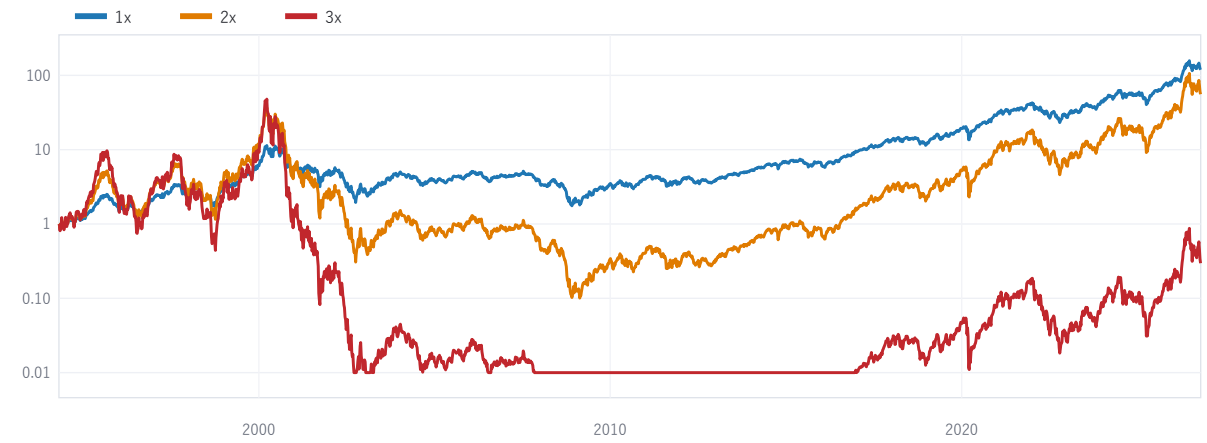


Figure 1. Growth of $1 (log scale)



Figure 2. Drawdown from peak

Reading the charts

The log scale makes compounding visible: the 1x line is close to straight, while the leveraged lines bend after every crash. The drawdown chart explains why. After the 2000-02 and 2008 episodes, the 3x series sits at the -100% floor, which means the position was effectively wiped out and could not participate in the recovery.

This is not a tail risk that can be waited out. Once the series reaches zero, the compounding stops permanently; the index can recover while the leveraged position cannot.

4. Decade-by-decade

Compounding over a full cycle hides the experience of any single decade. The table below shows the annualised return of each leverage level within each decade.

Period	1x	2x	3x
1994–1999	+38.1%	+55.1%	+50.2%
2000–2009	-5.8%	-30.3%	-58.1%
2010–2019	+18.6%	+31.5%	+38.5%
2020–2026	+30.3%	+40.7%	+30.1%

Leverage amplifies whichever regime you happen to invest in. A lost decade for the index becomes a catastrophic decade for 3x, and no holding period repairs it.

5. Rolling ten-year windows and sensitivity

Each point is the annualised return of a ten-year holding period starting on that day.

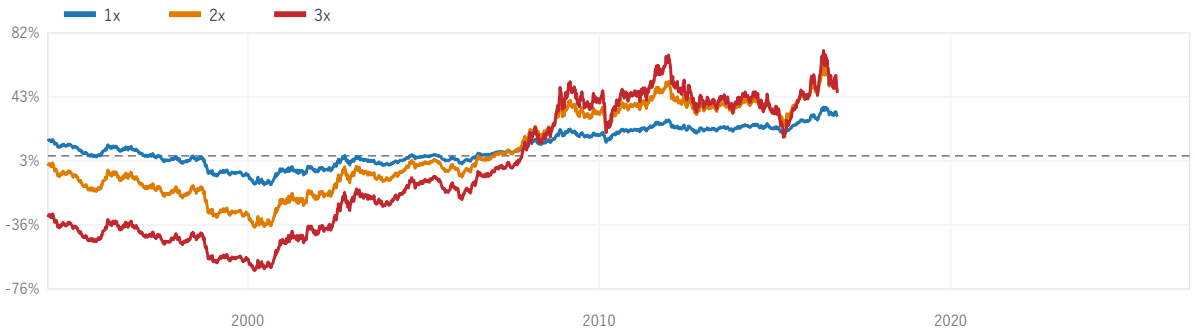


Figure 3. Rolling 10-year CAGR

Leverage	Worst	Median	Best	Share below zero
1x	-11.8%	+9.0%	+36.6%	16.1%
2x	-38.3%	+0.7%	+61.5%	48.9%
3x	-64.8%	-11.3%	+71.0%	58.3%

The 'share below zero' column is the practical risk of leverage. It is not a tail scenario; it is a repeated historical outcome.

6. Cross-index comparison

The same method applied to three indices with different volatility.

Index	Period	1x CAGR	2x CAGR	3x CAGR	3x max DD	3x Calmar
S&P 500	56y	+10.92%	+13.42%	+13.36%	97.9%	0.14
Nasdaq-100	41y	+15.29%	+19.81%	+17.25%	100.0%	0.17
SOX	32y	+15.87%	+13.21%	-3.63%	100.0%	-0.04

As volatility rises, the optimal leverage falls: 2x for the S&P 500 and Nasdaq-100, and 1x for SOX. In SOX, 3x lost money over 32 years despite the index compounding at 15.9% a year.

7. Limitations

The index excludes dividends; a constant yield is assumed. Real ETFs carry tracking error, spreads and taxes. The synthetic series is a single path and does not model intraday gaps, so tail risk is understated. Historical volatility is not a forecast.

Reproduction

All figures are generated by the Python scripts in the project. The seed is fixed and no external library is required beyond ReportLab for typesetting.

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